EE2004: Digital Signal Processing

Tutorial 9

(April 9, 2026)

1. Let $H(e^{j\omega}) = \sum_{n=0}^{N-1} h[n]\, e^{-j\omega n}$ where $h[n] = h_R[n] + j\, h_I[n]$, i.e., the filter coefficients $h[n]$ are complex-valued. It is given that the filter is linear phase. Derive the condition on the impulse response imposed by this constraint.

2. Both $H_1(e^{j\omega})$ and $H_2(e^{j\omega})$ are generalized linear phase systems. Which of the following systems must also belong to the same class? (a) $H_1(e^{j\omega}) + H_2(e^{j\omega})$, and (b) $H_1(e^{j\omega}) \cdot H_2(e^{j\omega})$.

3. A Type I FIR filter with real-valued coefficients has impulse response length $N = 9$. Its known zeros are $z_1 = -0.5$, $z_2 = 0.3 + j0.5$, and $z_3 = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$. Determine the location of the remaining zeros and find the expression for $H(z)$.

4. Let $H(z)$ be a Type I linear phase lowpass filter of length $N(= M-1)$; $H_{ZP}(\omega) \approx 1$ for $0 \le \omega \le \pi/4$, and $H_{ZP}(\omega) \approx 0$ for $\pi/3 \le \omega \le \pi$. Another filter $G(z)$ is obtained from $H(z)$ as $G(z) = z^{-M/2} - H(z)$. [Note: The notation $H_{ZP}(\omega)$ denotes the zero-phase response.]
   (a) Express the impulse response $g[n]$ in terms of $h[n]$. To what linear phase filter type does $g[n]$ belong?
   (b) Obtain an expression for $G_{ZP}(\omega)$ in terms of $H_{ZP}(\omega)$, and give a rough sketch of $G_{ZP}(\omega)$ for $0 \le \omega \le \pi$.
   (c) Compare the behaviour of $G_{ZP}(\omega)$ with that of $F_{ZP}(\omega)$ where $F(z)$ equals $(-1)^{M/2}H(-z)$.

5. Derive the closed-form expressions for $H_{ZP}(\omega)$ if $H(z)$ equals (a) $(1+z^{-1})^2(1-z^{-1})$, and (b) $(1+z^{-2})(1-z^{-1})^2$. For both these cases, derive and sketch $\angle H(e^{j\omega})$ over the interval $0 \le \omega \le \pi$.

6. Design a Type III linear phase filter with $M = 4$ satisfying the following conditions: $H_{ZP}(0.3\pi) = 0.2$ and $H_{ZP}(0.6\pi) = 0.8$.

7. Let $h[n]$ be the impulse response of a causal, linear phase FIR filter. We are given the following facts:
   (a) $\angle H(e^{j0+}) = -3\omega + \pi/2$.
   (b) $H(e^{j\pi}) = 0$.
   (c) $\frac{1}{2\pi}\int_{-\pi}^{\pi} |H(e^{j\omega})|^2\, d\omega = 28$.
   (d) Let $y[n] = h[n] * u[n]$. $y[1] = 4$ and $y[2] = 6$.
   (e) $\frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{Re}\{H(e^{j\omega})\}\, e^{j6\omega}\, d\omega = -1.5$.

   Find $h[n]$.